

ALIN KUMAR

Aspiring Data Scientist | Machine Learning | Data Analytics

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GitHub Profile | LinkedIn Profile

PROFESSIONAL SUMMARY

Aspiring Data Scientist with a strong foundation in Python, SQL, Machine Learning, and Data Analytics. Skilled in data preprocessing, exploratory data analysis, feature engineering, predictive modeling, and data visualization. Experienced in building end-to-end machine learning projects across financial fraud detection, telecom customer churn prediction, and lead conversion analytics while transforming complex data into actionable business insights.

TECHNICAL SKILLS

Languages:	Python, SQL
Machine Learning:	Scikit-learn, XGBoost, Random Forest, Logistic Regression, Decision Trees, KNN, Ensemble Learning, Hyperparameter Tuning, Cross Validation
Data Analysis:	Pandas, NumPy, Data Cleaning, Exploratory Data Analysis, Feature Engineering, Feature Selection, Statistical Analysis
Visualization:	Power BI, Matplotlib, Seaborn, Plotly
Deep Learning:	TensorFlow, Keras, EfficientNetBo, Focal Loss, Integrated Gradients
Tools & Technologies:	Git & GitHub, Google Colab, Jupyter Notebook, VS Code, MySQL, Microsoft Excel

PROJECTS

AI Industrial Steel – Defect Detection (Deep Learning) GitHub Repository: [GitHub Repository](#) | Live Demo: [Live Demo](#)

- Developed a deep learning model for defect detection in steel surface images.
- Achieved **87.86% Test Accuracy** and **79.61% Macro F1-Score** on 10-class classification.
- Used EfficientNet-Bo with transfer learning and balanced training for robust performance.
- Integrated Grad-CAM for visual explainability of defect predictions.

Tech Stack: Python, TensorFlow, Keras, EfficientNet-Bo, Focal Loss, Grad-CAM, Streamlit

Classes: crazing, inclusion, patches, pitted_surface, rolled-in_scale, scratches, scab, weld_bead, oil_spot, dirtiness

AI-Powered Financial Fraud Risk Intelligence System GitHub Repository: [GitHub Repository](#) | Live Demo: [Live Demo](#)

- Developed an AI-powered fraud detection pipeline using ensemble machine learning techniques.
- Achieved **99.98% Accuracy** and **99.56% ROC-AUC** on highly imbalanced financial transaction data.
- Improved fraud identification with **98.9% Precision** through feature engineering and model optimization.
- Integrated SHAP explainability to provide transparent and interpretable model predictions.

Tech Stack: Python, Pandas, Scikit-learn, XGBoost, SHAP, SQL

Brain Tumor Detection (Deep Learning) GitHub Repository: [GitHub Repository](#) | Live Demo: [Live Demo](#)

- Built a deep learning model for brain tumor classification using EfficientNetBo.
- Achieved **84.32% Validation Accuracy** during model evaluation.
- Used Focal Loss to handle class imbalance and improve minority class detection.
- Used Integrated Gradients for model explainability and attribution visualization.

Tech Stack: Python, TensorFlow, Keras, EfficientNetBo, Focal Loss, Integrated Gradients, Streamlit

Classes: Glioma, Meningioma, Pituitary, No Tumor

Lead Conversion Intelligence Platform GitHub Repository: [GitHub Repository](#) | Live Demo: [Live Demo](#)

- Developed a machine learning pipeline for lead conversion prediction.
- Achieved **87.1% Accuracy** with **80.8% Precision** after hyperparameter optimization.
- Automated lead prioritization using probability-based conversion scoring.
- Improved marketing decision-making through predictive analytics.

Tech Stack: Python, Pandas, Scikit-learn, SQL

EDUCATION

B.Sc. (Hons.) Computer Science

School of Open Learning (SOL)

University of Delhi

Pursuing

LINKS & PORTFOLIO

GitHub: [GitHub Profile](#)

LinkedIn: [LinkedIn Profile](#)

KEY STRENGTHS

Problem Solving | Analytical Thinking | Data-Driven Mindset | Communication | Continuous Learning | Team Collaboration